

Haixiang Lan

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Education

Columbia University

Ph.D. in Operations Research (Specialization: Data Science)

M.S. in Operations Research

Advisors: Adam Elmachtoub, Henry Lam

- GPA: 4.1/4

The Chinese University of Hong Kong, Shenzhen (CUHK(SZ))

B.S. in Applied Mathematics

- GPA: 3.94/4

University of Cambridge

Visiting Student

New York, United States

Aug. 2023 - Current

Aug. 2023 - Jun. 2024

Shenzhen, China

Sept. 2019 - May 2023

Virtual

Jul. 2021 - Aug. 2021

Publications and Preprints

- **Haixiang Lan**, Luofeng Liao, Adam Elmachtoub, Christian Kroer, Henry Lam, Haofeng Zhang. The Bias-Variance Tradeoff in Data-Driven Optimization: A Local Misspecification Perspective. In *NeurIPS'25: Proceedings of the 39th Annual Conference on Neural Information Processing Systems*, 2025.
- Adam Elmachtoub, Henry Lam, **Haixiang Lan**, Haofeng Zhang (α - β). Dissecting the Impact of Model Misspecification of Data-Driven Optimization. In *AISTATS'25: Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, 2025.
- **Haixiang Lan**, Guillermo Gallego, Zizhuo Wang, Yinyu Ye. LP-based Control for Network Revenue Management under Markovian Demands. In *WINE'24: Proceedings of the 20th Conference on Web and Internet Economics*, 2024.
- **Haixiang Lan**, Yinjun Wang, Yinyu Ye (α - β). A Tuning-Free Primal-Dual Splitting Algorithm for Large-Scale Semidefinite Programming. *INFORMS Optimization Society Conference (IOS)* 2024.
- Adam Elmachtoub, Rachitesh Kumar, **Haixiang Lan** (α - β). Achieving Low Regret for Online Contextual Allocation Problems. Submitted.
- Adam Elmachtoub, Rachitesh Kumar, **Haixiang Lan** (α - β). Data-Driven Admission Control under Cost Uncertainty. To be submitted.
- Adam Elmachtoub, Rachitesh Kumar, **Haixiang Lan** (α - β). Data-Driven Robust Group Selection. Working Paper.
- Henry Lam, **Haixiang Lan**, Tianyu Wang (α - β). Pretrain-Then-Calibrate: Reliable Contextual Optimization Along a Solution Path. To be Submitted.

α - β : Authors are listed in alphabetical order.

Presentations

- Data-Driven Admission Control under Cost-Uncertainty. *INFORMS Annual Meeting (November, 2026)*
- The Bias-Variance Tradeoff in Data-Driven Optimization: A Local Misspecification Perspective. *The 39th Annual Conference on Neural Information Processing Systems (December, 2025), Contextual Stochastic Optimization Workshop, (February, 2026)*.
- Dissecting the Impact of Model Misspecification of Data-Driven Optimization. *The 28th International Conference on Artificial Intelligence and Statistics (May, 2025), INFORMS Annual Meeting (October 2025), Concordia University (May, 2026)*
- LP-based Control for Network Revenue Management under Markovian Demands. *INFORMS Revenue Management and Pricing Section Conference (July 2024), INFORMS Annual Meeting (October 2024), The 20th Conference on Web and Internet Economics (December, 2024)*.
- A Tuning-Free Primal-Dual Splitting Algorithm for Large-Scale Semidefinite Programming. *INFORMS Opti-*

Industry Experience

Amazon

New York, United States

Applied Scientist Intern

Summer 2025

- Worked on *Long-Run Marginal Benefit Analysis for Real Estate Planning*.
- Developed a stochastic dynamic programming and mixed integer programming framework for the next-generation coordinated capacity and topology planning platform of Amazon fulfillment network.
- Implemented approaches to drive multi-billion dollar investment decisions shaping Amazon's global strategy.

Amazon

New York, United States

Applied Scientist Intern

Summer 2026

- Worked on *Accelerating Amazon's Strategic Planning Engine*.
- Developed (1) a temporal decomposition framework using distributed alternating direction method of multipliers (ADMM) and (2) a GPU-accelerated primal-dual hybrid gradient (PDHG) framework for Amazon's multi-year multi-commodity inventory flow planning problems.
- Implemented a code package for problem-solving acceleration, and achieved a $20\times$ running time reduction.

Honours and Awards

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| ◦ NeurIPS Scholar Award | 2025 |
| ◦ Cornelius A. Boyle Fellowship (Columbia IEOR Department Fellowship) | 2023 |
| ◦ Presidential Award in CUHK(SZ) (highest award for undergraduates) | 2023 |
| ◦ Academic Performance Scholarship (Class A) | 2019-2022 |
| ◦ Outstanding Undergraduate Student Teaching Fellow | 2023 |
| ◦ Half Tuition and Bowen Scholarship | 2019-2023 |
| ◦ Dean's list | 2019-2023 |
| ◦ Contemporary Undergraduate Mathematical Contest in Modeling: National Second Prize | 2021 |
| ◦ Undergraduate Research Award | 2021 |

Professional Service

- Reviewer for ICML 2026, NeurIPS 2025, 2026, TMLR.

Teaching Experience

At **Columbia University**: Teaching Assistant

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| ◦ IEOR 4106, Stochastic Models | Spring 2024 |
| ◦ IEOR 3106, Stochastic Systems and Applications | Fall 2024 |
| ◦ IEOR 4650, Business Analytics | Spring 2025, Fall 2025 |

At **CUHK(SZ)**: Undergraduate Student Teaching Fellow

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| ◦ CHM 1001, General Chemistry | Fall 2020 |
| ◦ MAT 1012, Honours Calculus II | Spring 2021 |
| ◦ MAT 2006, Elementary Real Analysis I | Fall 2021 |
| ◦ MAT 4001, Numerical Analysis | Fall 2022, Spring 2023 |
| ◦ DDA 3005, Numerical Methods | Fall 2022 |
| ◦ MAT 3007, Optimization I | Spring 2023 |

Skills

Languages: English (Fluent, TOEFL 107, GRE 334), Mandarin (Native)

Computer Languages: Python, R, Matlab, Julia, Gurobi, Xpress, Mosel, COPT, $\text{\LaTeX}$